Technical Report: Complete Business Analysis Project

Week 8 Capstone Project - Real World Business Analysis

This report documents the end-to-end data analysis workflow, from business framing and data preparation to EDA, statistical testing, churn modeling, insights, and implementation planning.

Dataset and Quality

The main analysis uses customer_churn.csv with 500 cleaned rows. Supporting datasets include sales_data.csv and house_prices.csv for portfolio breadth.

Metric	Value
Rows	500
Missing values after cleaning	0
Duplicate customers	0
Churn rate	10.6%

Analysis Techniques

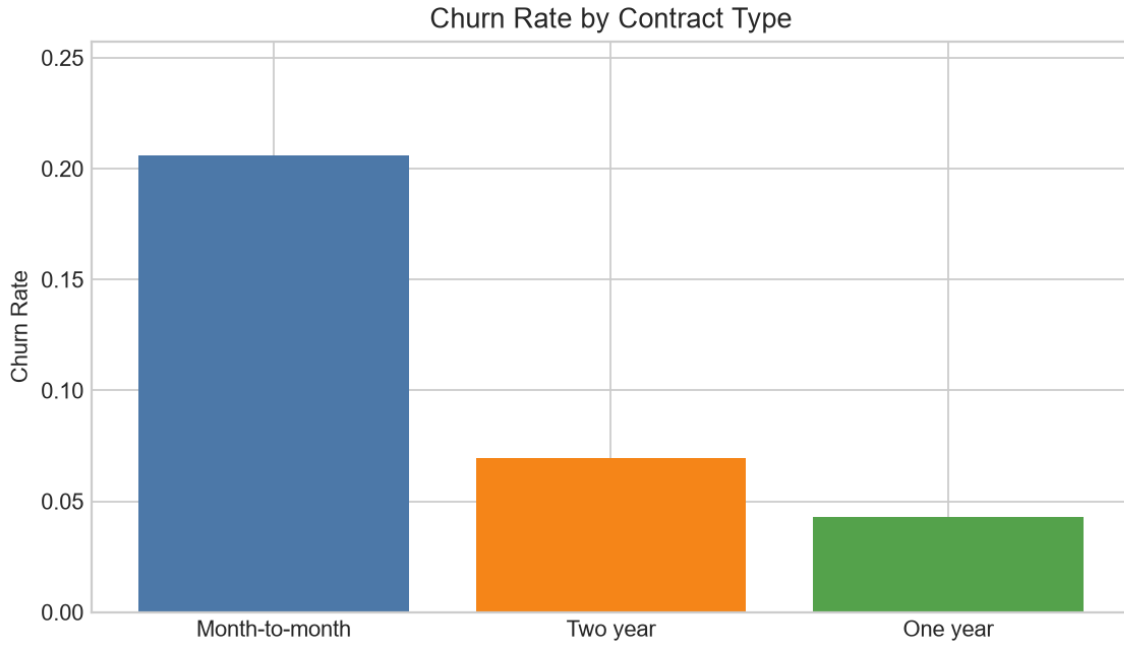
- Exploratory data analysis and segmentation
- Hypothesis testing with t-tests and chi-square test
- Correlation analysis
- Logistic regression churn modeling
- Business implementation planning

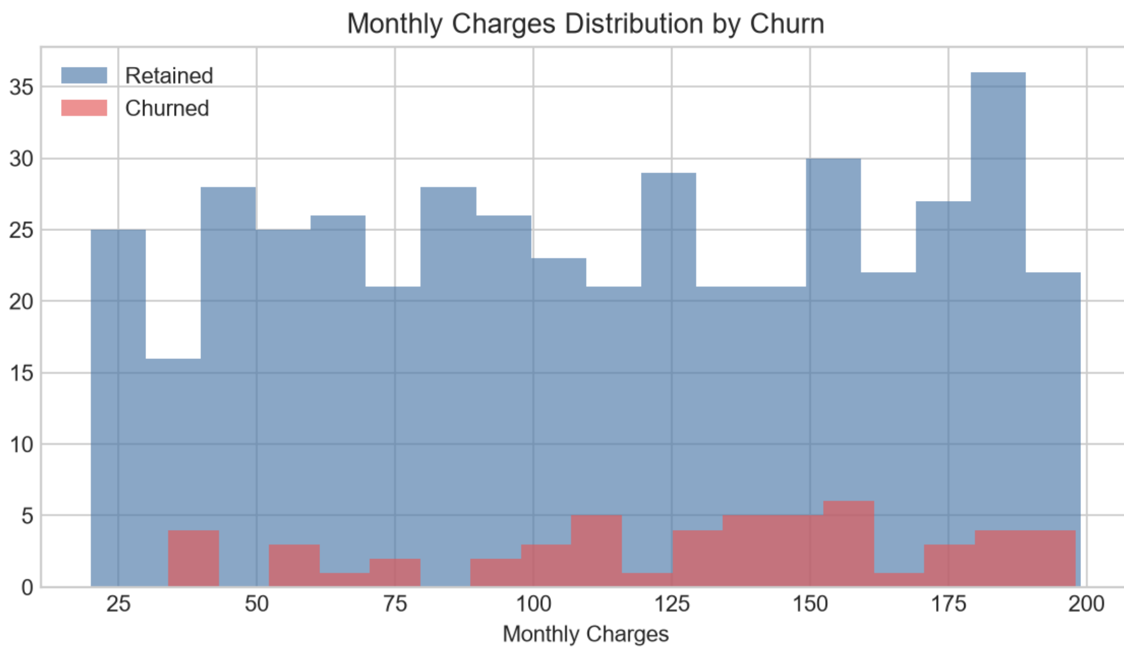
Statistical Findings

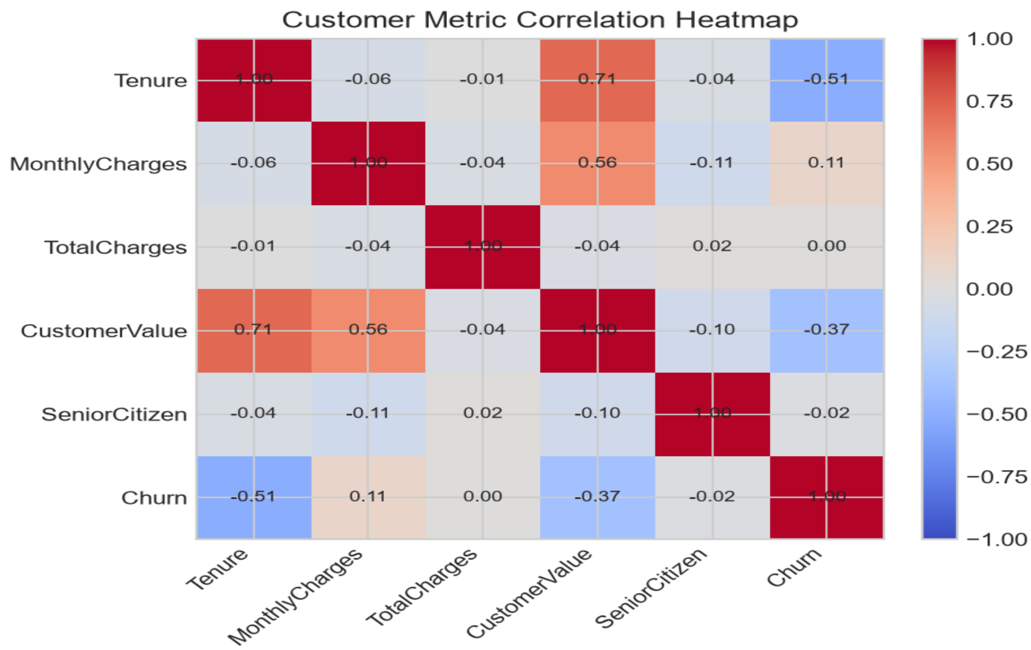
Tenure t-test p-value: 0.0000. Monthly charge t-test p-value: 0.0101. Contract/churn chi-square p-value: 0.0000.

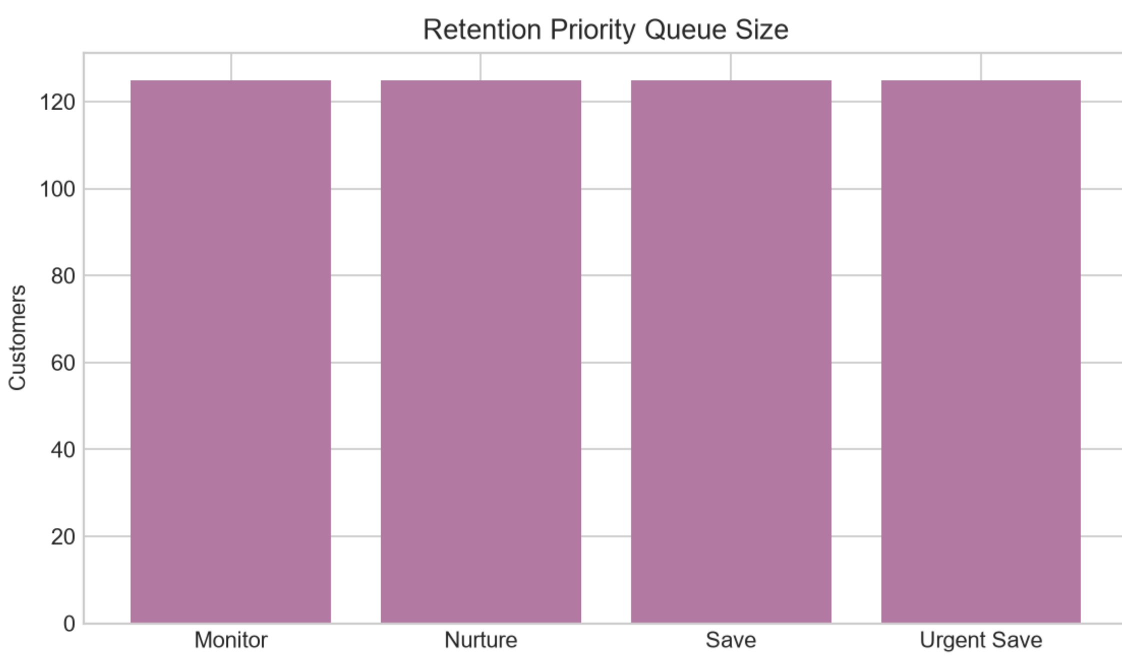
The logistic regression model was used to create a customer priority queue based on churn probability and revenue impact.

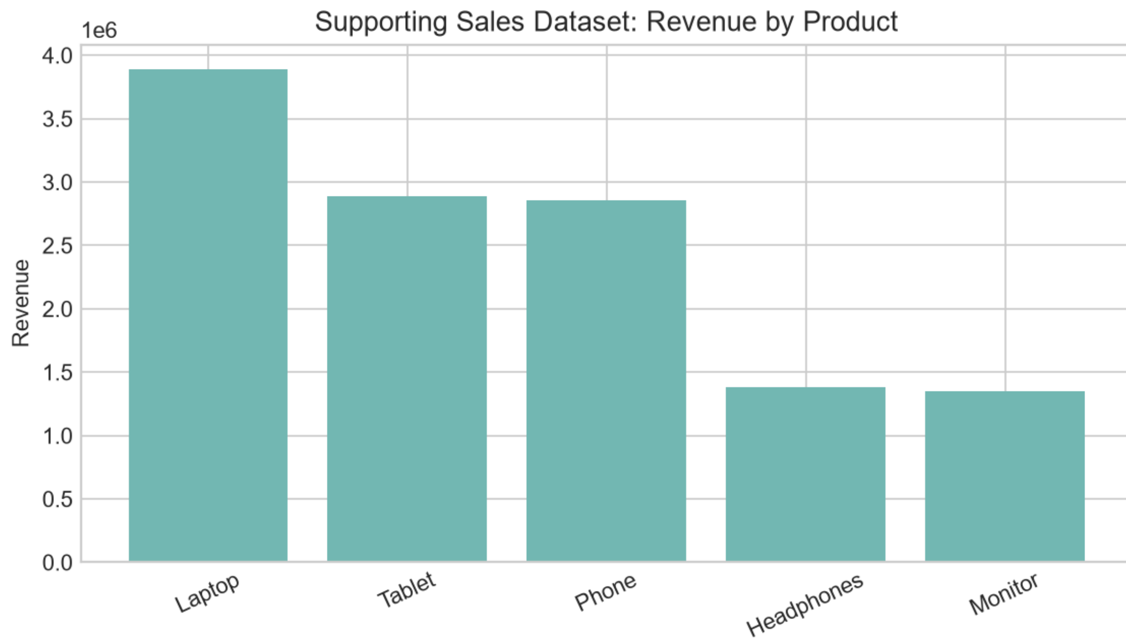
Visual Documentation



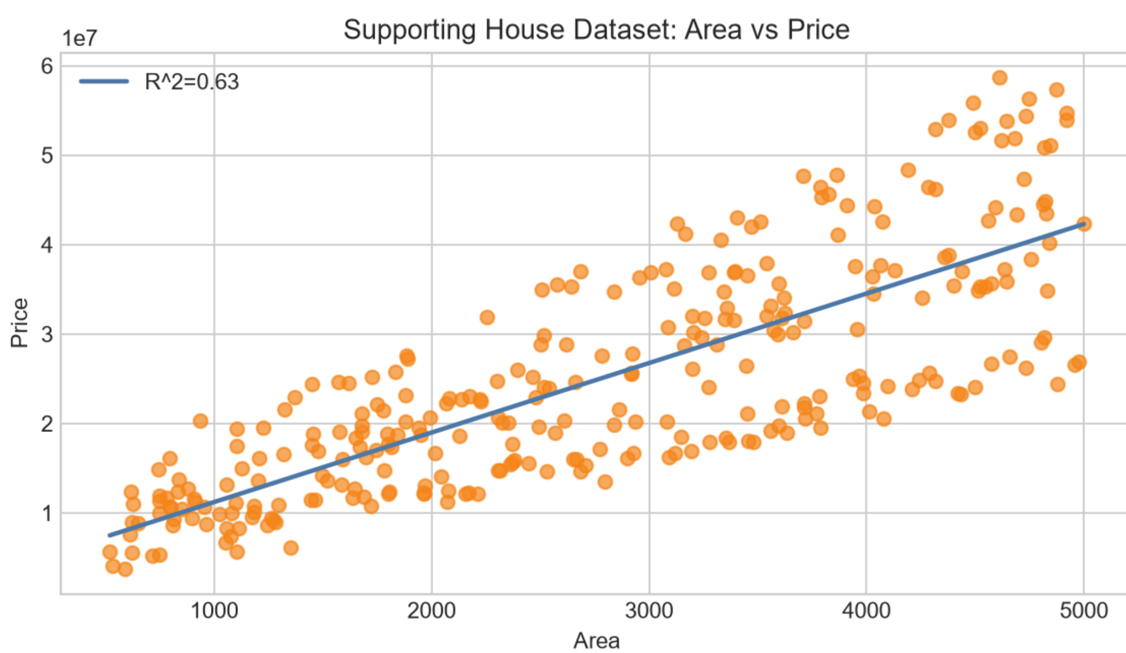



Customer Metric Correlation Heatmap
Tenure
MonthlyCharges
TotalCharges
CustomerValue
SeniorCitizen
Churn
1.00 -0.06 -0.01 0.71 -0.04 -0.51
-0.06 1.00 -0.04 0.56 -0.11 0.11
-0.01 -0.04 1.00 -0.04 0.02 0.00
0.71 0.56 -0.04 1.00 -0.10 -0.37
-0.04 -0.11 0.02 -0.10 1.00 -0.02
-0.51 0.11 0.00 -0.37 -0.02 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Retention Priority Queue Size
Customers
0
20
40
60
80
100
120
Monitor
Nurture
Save
Urgent Save



Supporting Sales Dataset: Revenue by Product
1e6
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Revenue
Laptop
Tablet
Phone
Headphones
Monitor



Supporting House Dataset: Area vs Price
1e7
6
5
4
3
2
1
Price
R^2=0.63
1000
2000
3000
4000
5000
Area

Implementation Plan

- Week 1: deploy churn-risk scoring and retention queue.
- Week 2: contact urgent-save customers with contract migration offers.
- Week 3: measure saved revenue, conversion rate, and churn reduction.
- Week 4: refine model using campaign response data.

Recommendations

- Prioritize month-to-month and electronic-check customers for proactive retention outreach.
- Build an urgent-save queue using predicted churn risk multiplied by monthly revenue impact.
- Offer annual contract migration incentives to customers with high monthly charges and short tenure.
- Track retention success with churn rate, saved monthly recurring revenue, and conversion to longer contracts.
- Use the same analysis pattern as a portfolio template for sales and real-estate business datasets.